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Electric energy storage device and electric tool system

Abstract

An electric energy storage device which includes four energy units with a substantially same voltage value is provided. Each energy unit is provided with a positive electrode and a negative electrode. The electric energy storage device comprises a socket with eight independently arranged electrode terminals that are connected with the four energy units. The disclosure also provides an electric tool system using the electric energy storage device. The electric tool is provided with plugs that may be connected with the four energy units in different states, allowing the electric energy storage device to output multiple voltages.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of U.S. patent application Ser. No. 17/344,940 filed Jun. 10, 2021, U.S. Ser. No. 17/344,940 being a continuation application of PCT/CN2019/114230 filed Oct. 30, 2019. PCT/CN2019/114230 is related to and claims the benefit of priority of the following commonly-owned, presently-pending Chinese patent applications: serial No. 201811564074.2, No. 201811564236.2, No. 201811564279.0, No. 201811566089.2, No. 201822145919.6, No. 201822145933.6, No. 201822146566.1, No. 201822146585.4, No. 201822146597.7, all filed Dec. 20, 2018, which the present application is a non-provisional application thereof. The disclosures of the forgoing applications are hereby incorporated by reference in their entirety, including any appendices or attachments thereof, for all purposes.

TECHNICAL FIELD

(1) The disclosure relates to the field of power tools, in particular to an electric energy storage device and an electric tool system.

BACKGROUND

(2) In the garden machinery and power tool industry, power tools are generally driven by rechargeable batteries, typically batteries consisting of a plurality of battery, and motors are driven by electrical energy stored in the batteries. After use, the voltage is reduced, the battery pack can be detached from the electric tool, and the external charging device is used for charging the battery pack.

(3) The electric tool has a rated working voltage, different electric tools have different rated working voltages, and the output voltage of the battery pack is fixed, so that different battery packs need to be prepared to adapt to the electric tools with different rated working voltages, which the operation is complicated, the cost is increased, and the resource is wasteful.

(4) Therefore, it is necessary to design an electric energy storage device and an electric tool system to solve the above problems.

SUMMARY

(5) This disclosure provides an electric energy storage device which can provide three output voltages, and an electric tool system using the electric energy storage device.

(6) To achieve the above object, the disclosure provides the following solution, an electric energy storage device which comprises four energy units with a substantially same voltage value, wherein each energy unit is provided with a positive electrode and a negative electrode, the electric energy storage device comprises a socket, and the socket comprises eight independently arranged electrode terminals; the eight electrode terminals comprise four positive terminals respectively connected with the positive poles of the four energy units and four negative terminals respectively connected with the negative poles of the four energy units.

(7) In an embodiment, each electrode terminal is provided with a contact part formed by two arc-shaped elastic arms, and the socket comprises a communication terminal and a charging terminal.

(8) In an embodiment, eight of the electrode terminals are evenly distributed into an upper row and a lower row, the polarities of the two electrode terminals corresponding to the upper position and the lower position are the same, and the corresponding two contact parts are vertically aligned.

(9) In an embodiment, four electrode terminals located at two outer sides are correspondingly connected with positive electrodes and negative electrodes of two energy units respectively, and four electrode terminals located at the inner sides are correspondingly connected with positive electrodes and negative electrodes of the other two energy units respectively.

(10) In an embodiment, the four electrode terminals in the upper row are a fourth negative terminal, a second negative terminal, a third positive terminal, and a first positive terminal, and the four electrode terminals in the lower row are a first negative terminal, a third negative terminal, a second positive terminal, and a fourth positive terminal.

(11) In an embodiment, eight of the electrode terminals are arranged in a row and sequentially comprise a fourth negative terminal, a third negative terminal, a second negative terminal, a first negative terminal, a second positive terminal, a third positive terminal, a fourth positive terminal, and a first positive terminal.

(12) To achieve the above object, the disclosure provides the following technical solution. An electric tool system includes an electric tool and the above electric energy storage device which is provided with a plug matched with the socket, the plug is provided with a plurality of male terminals, the plurality of male terminals are electrically connected with eight electrode terminals, and the four energy units form one of the following four connection states: a. four energy units are connected in parallel, b. each two of the four energy units are parallel connected first and then series connected therebetween, c. each two of the four energy units are series connected first and

then parallel connected therebetween, and d. four energy units are connected in series.

(13) In an embodiment, the plug includes a first male blade and a second male blade, the first male blade is connected to the four negative terminals of the four energy units, and the second male blade is connected to the four negative terminals of the four energy units.

(14) In an embodiment, the first male blade and the second male blade are each provided with two contact arms, wherein each of the contact arms is in contact with two electrode terminals; or the first male blade and the second male blade are each provided with four contact arms; each of the contact arms is in contact with one of the electrode terminals.

(15) In an embodiment, the plug includes a first male blade, a second male blade, and a third male blade, the first male blade is connected to the two negative terminals corresponding to two of the energy units, and the second male blade is connected to the two positive terminals corresponding to the other two energy units, and the third male blade is connected to the remaining four electrode terminals.

(16) In an embodiment, the first male blade and the second male blade are each provided with two contact arms, and the third male blade is provided with four contact arms.

(17) In an embodiment, the four contact arms of the third male blade are located in a row.

(18) In an embodiment, the plug comprises a first male blade, a second male blade, a third male blade, a fourth male blade, and a fifth male blade, wherein the first male blade is connected with a negative terminal corresponding to one energy unit, the fifth male blade is connected with a positive terminal corresponding to the other energy unit, and the second male blade, the third male blade and the fourth male blade are connected with the remaining six electrode terminals in series.

(19) In an embodiment, the first male blade and the fifth male blade are each provided with one contact arm, and the second male blade, the third male blade, and the fourth male blade each is provided with two contact arms respectively connected to different electrodes.

(20) The disclosure also provides the following technical solution. An electric tool system includes a low-voltage electric tool, a medium-voltage electric tool, and a high-voltage electric tool. The electric tool system further comprises the above electric energy storage device, the low-voltage electric tool is provided with a low-voltage plug, the low-voltage plug cooperates with the socket and the four energy units are parallel connected; the high-voltage electric tool is provided with a high-voltage plug which cooperates with the socket and enables the four energy units are series connected; the medium-voltage electric tool is provided with a medium-voltage plug, and the medium-voltage plug cooperates with the socket and enables the four energy units are connected in a medium-voltage state to provide a medium-voltage.

(21) The beneficial effects of the disclosure are: the electric energy storage device has a variety of output voltages, which increases the application range of the electric energy storage device and reduces the cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a parallel circuit diagram of four energy units of the disclosure.

(2) FIG. 2 is a circuit diagram showing each two of the four energy units connected in parallel and then connected in series therebetween of the disclosure.

(3) FIG. 3 is a circuit diagram showing each two of the four energy units connected in series and then connected in parallel therebetween according to the disclosure.

(4) FIG. 4 is a circuit diagram of four energy units in series in the disclosure.

(5) FIG. 5 is a perspective view of an electric energy storage device in accordance with the first embodiment of the disclosure.

(6) FIG. 6 is a plug-in view of a low voltage plug and a conductive terminal according to the first

embodiment of the disclosure.

(7) FIG. 7 is a block diagram of the low voltage plug of FIG. 6.

(8) FIG. 8 is a plug-in view of a medium voltage plug and a conductive terminal according to the first embodiment of the disclosure.

(9) FIG. 9 is a structural view of the medium voltage plug of FIG. 8.

(10) FIG. 10 is a block diagram of one of the medium voltage male blades of FIG. 9.

(11) FIG. 11 is a plug-in view of a high voltage plug and a conductive terminal according to the first embodiment of the disclosure.

(12) FIG. 12 is a structural view of the high voltage plug of FIG. 11.

(13) FIG. 13 is a block diagram of two of the high voltage male blades of FIG. 12.

(14) FIG. 14 is a perspective view of an electric energy storage device according to the second embodiment of the disclosure.

(15) FIG. 15 is a plug-in view of a low-voltage plug and a conductive terminal according to the second embodiment of the disclosure.

(16) FIG. 16 is a block diagram of the low voltage plug of FIG. 15.

(17) FIG. 17 is a plug-in view of a medium voltage plug and a conductive terminal according to the second embodiment of the disclosure.

(18) FIG. 18 is a structural view of the medium voltage plug of FIG. 17.

(19) FIG. 19 is a plug-in view of a high voltage plug and a conductive terminal.

(20) FIG. 20 is a structural view of the high voltage plug of FIG. 19.

DETAILED DESCRIPTION

(21) In order to make objects, aspects, and advantages of the disclosure more apparent, the disclosure will describe in detail with the drawings and specific examples.

(22) It should be noted that, in the drawings, wherein only structural and/or process steps have been shown and described in detail that are pertinent to the disclosure, other details that are not pertinent to the disclosure have been omitted so as not to obscure the disclosure with unnecessary detail.

(23) Additionally, it should also be noted that the terms “comprises, comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

First Embodiment

(24) The disclosure provides an electric energy storage device **100** including four energy units. The voltage value of each energy unit may be substantially the same, meaning that the measured voltage value of each of the four energy units may be nV , or $n \pm 5\% V$. The four energy units can output three different voltages based on different configurations of series or parallel connections:

(25) As shown in FIG. 1, all four energy units are connected in parallel, and the electric energy storage device **100** may output a low voltage of nV ;

(26) As shown in FIG. 2, the four energy units are divided into two pairs, each pair including two energy units connected in parallel, and the two pairs may then be connected in series. Alternatively, as shown in FIG. 3, the four energy units are divided into two pairs, each pair including two energy units connected in series, and the two pairs may then be connected in parallel. In the above two cases, the electric energy storage device **100** may output a medium voltage of $2 \cdot nV$;

(27) As shown in FIG. 4, when all four energy units are connected in series, the electric energy storage device **100** may output a high voltage of $4 \cdot nV$.

(28) It should be noted that each of the energy units may be a single battery or other energy carrier, or an electrical combination of multiple batteries or energy carriers; the electric energy storage device includes, but is not limited to, a lithium battery, a nickel-hydrogen battery, cadmium nickel batteries, and/or other rechargeable batteries. When a battery is used as an energy unit, the electric energy storage device **100** may also be referred to as a battery pack.

(29) As shown in FIG. 5, the electric energy storage device **100** may include a housing (not shown) and a socket. Four energy units may be accommodated in the housing, and the socket is electrically connected with the four energy units. The socket **110** is disposed on one side of the electric energy storage device **100**, and contains a plug interface for connecting with a mating-plug of a power tool. The socket **110** comprises a body **106** and multiple conductive terminals mutually independent of each other and arranged in the body **106**.

(30) As shown in FIG. 6, the multiple conductive terminals comprise eight electrode terminals each of which is electrically connected to the positive or the negative electrodes of the four energy units. Specifically: the eight electrode terminals include a first positive terminal **111** and a first negative terminal **112** respectively connected to the positive and negative electrodes of the first energy unit; a second positive terminal **113** and a second negative terminal **114** respectively connected to the positive and negative electrodes of the second energy unit; a third positive terminal **115** and a third negative terminal **116** respectively connected to the positive and negative electrodes of the third energy unit; and a fourth positive terminal **117** and a fourth negative terminal **118** respectively connected to the positive and negative electrodes of the fourth energy unit.

(31) The conductive terminals may also include a communication terminal **107** and a charging terminal **108** arranged outside the eight electrode terminals. The communication terminal **107** and the charging terminal **108** need to be separately arranged on two opposite ends of the conductive terminals, so that damage to the electric energy storage device caused by high voltage during charging can be prevented, and the service life of the electric energy storage device can be prolonged.

(32) In this embodiment, the eight electrode terminals may be evenly distributed into two rows (an upper row and a lower row), and the communication terminal **107** and the charging terminal **108** are respectively arranged on two sides of the two rows of electrode terminals. That is, four electrode terminals located on the outer sides of the upper row and the lower row are connected with positive electrodes and negative electrodes of two energy units, the four electrode terminals located on the inner sides of the upper row and the lower row are connected with positive electrodes and negative electrodes of the other two energy units. Further, the two electrode terminals corresponding to the positive electrodes and the negative electrodes of the same energy unit are arranged in a diagonal relation. Specifically, the electrode terminals in the upper row are, from left to right, a fourth negative terminal **118**, a second negative terminal **114**, a third positive terminal **115**, and a first positive terminal **111**; the electrode terminals in the lower row are, from left to right, a first negative terminal **112**, a third negative terminal **116**, a second positive terminal **113**, and a fourth positive terminal **117**.

(33) Each of the electrode terminals **111**, **112**, **113**, **114**, **115**, **116**, **117**, **118** is provided with a contact part. The contact part comprises two arc-shaped elastic arms which protrude oppositely, and a vertical plugging gap is formed between the two elastic arms, in which a male blade of a connecting plug may be inserted. The polarity of the two electrode terminals corresponding to the positions in the upper row and the lower row may be the same, and the two contact parts are vertically aligned and can be contacted with the same connecting-plug male blade. For example, the fourth negative terminal **118** located at the first position of the upper left row may aligned with the contact part of the first negative terminal **112** located at the first position of the lower left row, and so on.

(34) The electric energy storage device **100** of the disclosure is capable of providing energy to a plurality of power tools having different rated voltages, each of the power tools having a corresponding plug that interfaces with a receptacle **110** of the electric energy storage device **100**. The plug may include an insulating base (not shown) and a plurality of male blades mounted on the insulating base (not shown). Different electric tools are provided with different plugs, and different plugs may be provided with male blades in different arrangement states, in order to generate different connection states with the eight electrode terminals of the socket, so that different

connection states are formed among the four energy units to form different output voltages. Each of the plugs further includes a communication male blade coupled with the communication terminal **107**, and a charging male blade coupled with the charging terminal **108**. The communication male blade and the charging male blade may not participate in voltage output, so no additional details are disclosed.

(35) FIG. **6** shows a situation in which the socket **110** of the electric energy storage device **100** can be configured to insert a low-voltage plug **120** (shown in FIG. **7**) of a low-voltage electric tool.

(36) As shown in FIG. **7**, the low-voltage plug **120** includes a communication male blade **124** and a charging male blade **125** located at the two outermost sides, in order to be mated respectively with the communication terminal **107** and the charging terminal **108**. The low-voltage plug **120** further includes two independently disposed low-voltage male blades **121** and **122**. Specifically, the male blade **121** may include two oppositely disposed contact arms **1211**, **1212**, a connection portion **1213** connects the two contact arms **1211** and **1212**, and a voltage output portion **1214** may be connected to the contact arm **1211** and the contact arm **1212**. The two low-voltage male blades **121** and **122** have a similar structure. Taking the low-voltage male blade **121** as an example, the contact arms **1211** and **1212** are located at the front end, and the voltage output part **1214** are respectively located at the rear end, of the low-voltage male blade **121**. The contact arms **1211** and **1212** and the voltage output part **1214** are vertically arranged, and the voltage output part **1214** and one of the contact arms **1211** and **1212** are located in the same vertical plane. The connection portion **1213** may horizontally connect the bottom edges of the two contact arms **1211**, **1212** near one side of the voltage output portion **1214** at the middle section of the low-voltage male blade **121**.

(37) Referring back to FIG. **6**, when the socket **110** is mated with the low-voltage plug **120**, each of the contact arms **1211** and **1212** (of the low-voltage male blades **121** and **122**) may be connected to a corresponding contact portion of two electrode terminals, which are up-down aligned.

Specifically, the contact arm **1211** of the low voltage male blade **121** may be plugged into the fourth negative terminal **118** and the first negative terminal **112**, and simultaneously the contact arm **1212** may be plugged into the second negative terminal **114** and the third negative terminal **116**. That is, the low voltage male blade **121** connects the negative electrodes of the four energy units together.

(38) The contact arm **1221** of the low voltage male blade **122** may be plugged into the third positive terminal **115** and the second positive terminal **113**, and simultaneously the contact arm **1222** may be plugged into the first positive terminal **111** and the fourth positive terminal **118**. That is, the low voltage male blade **122** connects the positive electrodes of the four energy units together. Thus, the low-voltage male blade **121** and the low-voltage male blade **122** may connect four energy units in parallel, and as shown in FIG. **1**, and the electric energy storage device **100** may output through two voltage outputs **1214**, **1224** a low voltage of nV for the low-voltage power tool.

(39) FIG. **8** shows a situation in which the socket **110** of the electric energy storage device **100** can be configured to insert a medium voltage plug **130** (shown in FIG. **9**) of a medium voltage electric tool.

(40) As shown in FIG. **9**, the medium voltage plug **130** includes a communication male blade (not labeled) and a charging male blade (not labeled) located on the two outermost sides, in order to respectively be mated and plugged into the communication terminal **107** and the charging terminal **108**. The medium voltage plug **130** includes three independently set medium voltage male blades **131**, **132**, **133**. The three medium voltage male blades **131**, **132**, **133** may be installed on an insulating base (not shown). The medium voltage male blade **131** may include two contact arms **1311**, **1312**, a connection portion **1313** connecting the two contact arms **1311**, **1312**, and a voltage output portion **1314** connected with the contact arms **1311** and **1312**. The medium voltage male blade **132** may include two contact arms **1321**, **1322** opposite to each other, a connection portion **1323** connecting the two contact arms **1321**, **1322**, and a voltage output portion **1324** connected to

the contact arms **1321** and **1322**. The two medium-voltage male blades **131**, **132** are arranged side by side.

(41) The structure of the medium-voltage male blades **131**, **132** may be substantially the same as that of the low-voltage male blades **121**, **122** of the low-voltage plug, except that the vertical height may be relatively low. That is, the male blades **131** and **132** may be connected to the four electrode terminals in the lower row of the socket **110**. As shown in FIG. **10**, the medium-voltage male blade **133** includes four contact arms **1331**, **1332**, **1333**, **1334** arranged side by side, and three connection portions **1335**, **1336**, **1337** connecting the adjacent contact arms **1331**, **1332**, **1333**, **1334** in pairs. The medium voltage male blade **133** are provided above the two medium voltage male blades **131**, **132** for connection with the four electrode terminals in the upper row of the socket **110**.

(42) Returning to FIG. **8**, when the socket **110** is mated with the medium voltage plug **130**, the contact arms **1311**, **1312** of the medium voltage male blade **131** mate respectively with the first and third negative terminals **112** and **116** which are in the lower row. The contact arms **1321**, **1322** of the medium voltage male blade **132** are respectively plugged into the second positive terminal **113** and the fourth positive terminal **117** located in the lower row. That is, the medium voltage male blade **131** connects two negative electrodes of the first and third energy units in parallel, and the medium voltage male blade **132** connects two positive electrodes of the second and fourth energy units in parallel.

(43) The contact arms **1331**, **1332**, **1333**, **1334** of the medium voltage male blade **133** are plugged into four electrode terminals located in the upper row. Specifically, the contact arms **1331**, **1332** and the connection piece **1335** connect the fourth negative terminal **118** and the second negative terminal **114** of the second group in parallel, and the contact arms **1333**, **1334** and the connection piece **1337** connect the third positive terminal **115** and the first positive terminal **111** in parallel. In combination with the connection of the two preceding medium voltage male blades **131**, **132** to the lower electrode terminals, the first and third energy units are already in parallel, and the second and fourth energy units are already in parallel. Finally, the first and third energy units (which are connected in parallel) and the second and fourth energy units (which are connected in parallel) are connected in series through the connection piece **1336** of the medium-voltage male blade **133**, so that each two of the four energy units are connected in parallel and then connected in series therebetween, as shown in FIG. **2**, the electric energy storage device **100** may output through the two voltage output parts **1314**, **1324** a medium voltage of $2 \times nV$ for the medium-voltage power tool.

(44) It can be seen in different embodiments, that the plug can be configured as a combination of male blades in various forms, such that when the plug is mated with the socket **110**, each two of the four energy units are connected in series and then connected in parallel therebetween to output a voltage of $2 \times nV$. Similarly, as shown in FIG. **3**, the energy units of FIG. **3** can be viewed respectively from top to bottom as first, third, second, and fourth units. The connection is compared with the connection according to the connection state in FIG. **3**, which will not be described in detail herein.

(45) FIG. **11** shows a situation in which the socket **110** of the electric energy storage device **100** can be configured to insert a high-voltage plug **140** (shown in FIG. **12**) of a high-voltage electric tool.

(46) As shown in FIG. **12**, the high-voltage plug **140** includes a communication male blade (not numbered) and a charging male blade (not numbered) which are located at the two outermost sides, in order to respectively be matched and plugged with the communication terminal **107** and the charging terminal **108**. The high voltage plug **140** includes five independently disposed high voltage male blades **141**, **142**, **143**, **144**, **145** mounted on a base (not shown). The high voltage male blades **142**, **143** are located in the upper row, the high voltage male blades **141**, **144**, **145** are located in the lower row, and the high voltage male blade **144** is located between the high voltage male blades **141**, **145**.

(47) As shown in conjunction with FIG. **12** and FIG. **13**, the high-voltage male blade **141** includes a contact arm **1411** to which the first negative terminal **112** is inserted, and a voltage output portion

1412 is connected to the contact arm **1411**. The high-voltage male blade **145** includes a contact arm **1451** and a voltage output part **1452** connected with the contact arm **1451**, and the contact arm **1451** is in contact with the fourth positive terminal **117**.

(48) The high voltage male blade **142** includes two oppositely disposed contact arms **1421**, **1422**, and includes a connection portion **1423** connecting the contact arms **1421**, **1422**. The high voltage male blade **143** includes two oppositely disposed contact arms **1431**, **1432**, and includes a connection portion **1433** connecting the contact arms **1431**, **1432**. The high voltage male blade **144** includes two oppositely disposed contact arms **1441**, **1442**, and includes a connection portion **1443** connecting the contact arms **1441**, **1442**. Contact arm **1421** is positioned directly above contact arm **1411**, contact arm **1422** is positioned directly above contact arm **1442**, contact arm **1431** is positioned directly above contact arm **1441**, and contact arm **1432** is positioned directly above contact arm **1451**.

(49) Returning to FIG. **11**, when the socket **110** is paired with the high voltage plug **140**, the contact arm **1411** of the high voltage male blade **141** is in connection with the first negative terminal **112**; the contact arm **1451** of the high voltage male blade **145** is in connection with the fourth positive terminal **117**. The contact arms **1421**, **1422** of the high voltage male blade **142** are respectively connected with the fourth negative terminal **118** and the third positive terminal **115**. That is, the positive electrode of the third energy unit and the negative electrode of the fourth energy unit are connected in series. The contact arms **1431**, **1432** of the high-voltage male blade **143** are respectively connected with the second negative terminal **114** and the first positive terminal **111**. That is, the positive electrode of the first energy unit and the negative electrode of the second energy unit are connected in series.

(50) The contact arms **1441**, **1442** of the high voltage male blade **144** are respectively connected with the third negative terminal **116** and the second positive terminal **113**. That is, the positive electrode of the second energy unit and the negative electrode of the third energy unit are connected in series. The high voltage male blades **142**, **143** and **144** connect four energy units in series. As shown in FIG. **4**, the electric energy storage device **100** outputs a voltage of $4 \times nV$ to the high voltage power tool through two voltage outputs **1412**, **1452**.

Second Embodiment

(51) As shown in FIG. **14**, which is another embodiment of the electric energy storage device, it differs from the electric energy storage device **100** of the first embodiment in the arrangement of the conductive terminals.

(52) The electric energy storage device **200** includes a housing (not shown), four energy units housed within the housing, and a socket **210** electrically connected to the four energy units. As shown in conjunction with FIG. **15**, the socket **210** includes a body **206** and multiple independently arranged conductive terminals housed in the body **206**. The multiple conductive terminals may include eight electrode terminals electrically connected to positive and negative electrodes of four energy units. Specifically, a first positive terminal **211** and a first negative terminal may be connected to positive and negative electrodes of a first energy unit **212**, a second positive terminal **213** and a second negative terminal **214** may be connected to the positive and negative electrodes of the second energy unit, a third positive terminal **215** and a third negative terminal **216** may be connected to the positive and negative electrodes of the third energy unit, and a fourth positive terminal **217** and a fourth negative terminal **218** may be connected to the positive and negative electrodes of the fourth energy unit. The multiple conductive terminals may further include a communication terminal and a charging terminal (not shown) in addition to the eight electrode terminals.

(53) In this embodiment, the eight electrode terminals may be arranged in a line, from left to right, as the fourth negative terminal **218**, the third negative terminal **216**, the second negative terminal **214**, the first negative terminal **212**, the second positive terminal **213**, the third positive terminal **215**, the fourth positive terminal **217**, and the first positive terminal **211**. The contact portion of

each electrode terminal is substantially the same as the contact portion referred to in the first embodiment.

(54) The electric energy storage device **200** of the disclosure is also capable of supplying energy to various power tools having different rated voltages, each power tool being provided with a plug which interfaces with the same socket **210** of the electric energy storage device **200**. The different plugging configurations may use different connection states established among the eight electrode terminals of the socket **210**. Specifically, the different connection states are formed among the four battery packs to generate different output voltages, which can be seen with reference to FIGS. **1** to **4**. Each plug comprises a communication male blade (not shown) matched and plugged with the communication terminal, and a charging male blade (not shown) matched and plugged with the charging terminal, which is for voltage input.

(55) FIG. **15** shows a situation where the conductive terminals of the socket **210** of the electric energy storage device **200** are matched with the low-voltage male blade of the low-voltage plug **220** of a low-voltage electric tool.

(56) Referring to FIG. **16**, the low-voltage plug **220** includes two independently disposed low-voltage male blades **221**, **222** mounted on an insulating base (not shown). The low-voltage male blade **221** includes four contact arms **2211**, **2212**, **2213**, **2214** arranged side by side, three connection portions **2215**, **2216**, **2217** each connecting two of the adjacent contact arms, and a voltage output part **2218** connecting with the contact arms **2211**, **2212**, **2213**, **2214**. The voltage output part **2218** is arranged on the rear side of the low-voltage male blade **221** and can be connected with one of the contact arms **2211**, **2212**, **2213**, **2214** lies in the same vertical plane. The low-voltage male blade **222** includes four contact arms **2221**, **2222**, **2223**, **2224** arranged side by side, three connection portions **2225**, **2226**, **2227** each connecting two of the adjacent contact arms, and a voltage output portion **2228** connecting to the contact arms **2221**, **2222**, **2223**, **2224**. The low-voltage male blade **222** may have a similar structure as the low-voltage male blade **221**.

(57) Referring back to FIG. **15**, when the socket **210** is mated with the low-voltage plug **220**, the contact arms **2211**, **2212**, **2213**, **2214** of the low-voltage male blade **221** are sequentially in connection with the fourth negative terminal **218**, the third negative terminal **216**, the second negative terminal **214**, and the first negative terminal **212**. That is, the contact arms **2211**, **2212**, **2213**, **2214** are connected in parallel with the negative electrodes of the four energy units. The contact arms **2221**, **2222**, **2223**, **2224** of the low-voltage male blade **222** are sequentially in connection with the second positive terminal **213**, the third positive terminal **215**, the fourth positive terminal **217**, and the first positive terminal **211**. That is, contact arms **2221**, **2222**, **2223**, **2224** are connected in parallel with the positive electrodes of the four energy units. Thus, two low-voltage male blades **221**, **222** are coupled in parallel to four energy units, and as shown in FIG. **1**, the electric energy storage device **200** may outputs an nV voltage for the low-voltage power tool through two voltage outputs **2218**, **2228**.

(58) FIG. **17** shows a situation in which the conductive terminals of the socket **210** of the electric energy storage device **200** are matched with the medium voltage male blades of the medium voltage plug **230** of a medium voltage electric tool.

(59) As shown in connection with FIG. **18**, the medium voltage plug **230** includes three independently arranged medium voltage male blades **231**, **232**, **233** positioned side by side, with the medium voltage male blades **231**, **232**, **233** being mounted on a base (not shown). The medium voltage male blade **231** includes two oppositely disposed contact arms **2311**, **2312**, a connection portion **2313** connecting the contact arms **2311**, **2312**, and a voltage output portion **2314** connecting to the contact arms **2311**, **2312**. The medium-voltage male blade **232** includes two oppositely arranged contact arms **2321** and **2322**, a connection portion **2323** connecting the contact arms **2321** and **2322** and a voltage output part **2324** connecting with the contact arms **2321** and **2322**. The medium-voltage male blade **233** includes four contact arms **2331**, **2332**, **2333**, **2334** arranged side by side, and three connection portions **2335**, **2336**, **2337** each connecting two

adjacent contact arms.

(60) Returning to FIG. 17, when the socket **210** is mated with the medium voltage plug **230**, the two contact arms **2311**, **2312** of the medium voltage male blade **231** are mated respectively with the fourth negative terminal **218** and the third negative terminal **216**. The two contact arms **2321**, **2322** of the medium voltage male blade **232** are respectively plugged with the second positive terminal **213** and the first positive terminal **211**. The four contact arms **2331**, **2332**, **2333**, **2334** of the medium voltage male blade **233** are sequentially plugged with the second negative terminal **214**, the first negative terminal **212**, the third positive terminal **215**, and the fourth positive terminal **217**. Thus, the first and second energy units are connected in parallel, and the third and fourth energy units are first connected in parallel and then connected in series with the first and second energy units (which are connected in parallel). As shown in FIG. 2, the electric energy storage device **200** outputs a voltage of $2 \times nV$ to the medium voltage power tool through the two voltage outputs **2314**, **2324**.

(61) It can be seen in other embodiments, the plug may be provided as a combination of other types of male blades so that when the plug is mated with the socket **210**, each two of the four energy units are connected in series and then connected in parallel therebetween to output a voltage of $2 \times nV$, which will not be described in detail herein.

(62) FIG. 19 shows a situation where the conductive terminals of the socket **210** of the electric energy storage device **200** are matched with the high-voltage male blades of the high-voltage plug **330** of a high-voltage electric tool.

(63) As shown in connection with FIG. 20, the high voltage plug **240** includes five independently disposed high voltage male blades **241**, **242**, **243**, **244**, **245** mounted on a base (not shown). Here, the high-voltage male blade **241** includes a contact arm **2411** and a voltage output portion **2412** connected to the contact arm **2411**. The high voltage male blade **245** includes a contact arm **2451** and a voltage output **2452** connected to the contact arm **2451**. The high voltage male blade **242** includes two contact arms **2421**, **2422** and a connection portion **2423** connecting the two contact arms **2421**, **2422**. The high-voltage male blade **243** includes two contact arms **2431**, **2432** and a connection portion **2433** connecting the contact arms **2431**, **2432**. The high voltage male blade **244** includes two contact arms **2441**, **2442** and a connection portion **2443** connecting the contact arms **2441**, **2442**.

(64) Returning to FIG. 19, when the socket **210** is mated with the high voltage plug **240**, the contact arm **2411** of the high voltage male blade **241** is mated with the fourth negative terminal **218**, the contact arm **2451** of the high voltage male blade **245** is mated with the first positive terminal **211**, and the contact arms **2421**, **2422** of the high voltage male blade **242** are mated with the fourth positive terminal **217** and the third negative terminal **216** respectively, so that the third and fourth energy units are connected in series. The contact arms **2431** and **2432** of the high-voltage male blade **243** are respectively plugged with the third positive terminal **215** and the second negative terminal **214**, so that the third and second energy units are connected in series. The contact arms **2441** and **2442** of the high-voltage male blade **244** are respectively plugged with the second positive terminal **213** and the first negative terminal **212**, so that the second and first energy units are connected in series. That is, the high-voltage male blades **241**, **242**, **243** are connected in series with four energy units, the contact arm **2411** of the high-voltage male blade **241** is inserted into the fourth negative terminal **218**, and the contact arm **2451** of the high-voltage male blade **245** is inserted into the first positive terminal **211**. As shown in FIG. 19, the electric energy storage device **100** outputs a voltage of $4 \times nV$ to the high-voltage power tool through the two voltage output portions **2412**, **2452**.

(65) In summary, four energy units with substantially the same voltage are respectively matched and plugged with a low-voltage plug, a medium-voltage plug, and a high-voltage plug through electrode terminals. Specifically, the four energy units may be mutually connected in parallel to output a low voltage; each two of the four energy units may be parallel connected first and then

series connected therebetween or each two of the four energy units may be series connected first and then parallel connected therebetween to output a medium voltage, or the four energy units may be mutually connected in series to output a high voltage. Therefore, the same electric energy storage device can selectively supply power to three electric tools with different rated working voltages. Such an operation is simple and convenient, and the cost is reduced.

(66) Therefore, although the disclosure has been described herein with reference to specific embodiments, freedom of modification, various changes and substitutions are intended to be included in the above disclosure, and it should be understood that, in some cases, on the premise of being without departing from the scope and spirit of the proposed disclosure, some features of the disclosure will be adopted without corresponding use of other features. Therefore, many modifications can be made to adapt specific environments or materials to the essential scope and spirit of the disclosure. The disclosure is not intended to be limited to specific terms used in the following claims and/or specific embodiments disclosed as best ways to implement the disclosure, but the disclosure will include any and all embodiments and equivalents falling within the scope of the appended claims. Therefore, the scope of the disclosure will only be determined by the appended claims.

Claims

1. A battery pack, comprising: a first string of battery cells; a second string of battery cells; a third string of battery cells; a fourth string of battery cells; a battery interface electrically connected to said first string of battery cells, said second string of battery cells, said third string of battery cells and said fourth string of battery cells; said battery interface is configured to output a first rated voltage, a second rated voltage, or a third rated voltage when a respective low voltage plug, a medium voltage plug or a high voltage plug is inserted in said battery interface; when the low voltage plug of a first power tool is inserted in said battery interface, said battery interface is electrically connected to the first power tool and triggered, said battery interface outputs the first rated voltage to said first power tool; when the medium voltage plug of a second power tool is inserted in said battery interface, said battery interface is electrically connected to the second power tool and triggered, said battery interface outputs the second rated voltage to said second power tool; when the high voltage plug of a third power tool is inserted in said battery interface, said battery interface is electrically connected to the third power tool and triggered, said battery interface outputs the third rated voltage to said third power tool; said first rated voltage, said second rated voltage, and said third rated voltage are all different; each string of the first, second, third and fourth strings of battery cells is provided with a positive electrode and a negative electrode; said battery interface comprises eight electrode terminals each of which is electrically connected to the positive or the negative electrodes of each string, different electric tools are provided with different plugs, and the different plugs are provided with male blades in different arrangement states, in order to generate different connection states with the eight electrode terminals of the said battery interface, so that the different connection states are formed among the first, second, third and fourth strings to output different voltages.
2. The battery pack according to claim 1, wherein said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells are connected in parallel when said battery interface is electrically connected to said first power tool.
3. The battery pack according to claim 1, wherein said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells are connected in series parallel when said battery interface is electrically connected to said second power tool.
4. The battery pack according to claim 1, wherein said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells are

connected in series when said battery interface is electrically connected to said third power tool.

5. The battery pack according to claim 1, wherein the rated voltage of each string of battery cells is approximately n V, the first rated voltage is n V, the second rated voltage is $2n$ V, the third rated voltage is $4n$ V.

6. The battery pack according to claim 5, wherein when n is 18, the first rated voltage is approximately 18V, the second rated voltage is approximately 36V, the third rated voltage is approximately 72V.

7. The battery pack according to claim 5, wherein when n is 20, the first rated voltage is approximately 20V, the second rated voltage is approximately 40V, the third rated voltage is approximately 80V.

8. The battery pack according to claim 1, wherein said battery interface comprises a plurality of outputs, said plurality of outputs being arranged in an array.

9. The battery pack according to claim 1, wherein said battery interface comprises a plurality of outputs, said plurality of outputs being configured to be arranged in upper and lower rows.

10. The battery pack according to claim 1, wherein said battery interface comprises a fourth positive output terminal, a fourth negative output terminal, a third positive output terminal, a third negative output terminal, a second positive output terminal, a second negative output terminal, a first positive output terminal and a first negative output terminal.

11. A battery pack comprising: a plurality of strings of battery cells; a battery interface electrically connected to said plurality of strings of battery cells; said battery interface is configured to output a first rated voltage, a second rated voltage, or a third rated voltage when a respective low voltage plug, a medium voltage plug or a high voltage plug is inserted in said battery interface; when the low voltage plug of a first power tool is inserted in said battery interface, said battery interface is electrically connected to a first power tool, said plurality of strings of battery cells output the first rated voltage outwardly in a first connection; when the medium voltage plug of a second power tool is inserted in said battery interface, said battery interface is electrically connected to a second power tool, said plurality of strings of battery cells output the second rated voltage outwardly in a second connection; when the high voltage plug of a third power tool is inserted in said battery interface, said battery interface is electrically connected to a third power tool, said plurality of strings of battery cells output the third rated voltage outwardly in a third connection; said first rated voltage being less than said second rated voltage and said second rated voltage being less than said third rated voltage; each string of the first, second, third and fourth strings of battery cells is provided with a positive electrode and a negative electrode; said battery interface comprises eight electrode terminals each of which is electrically connected to the positive or the negative electrodes of each string, different electric tools are provided with different plugs, and the different plugs is provided with male blades in different arrangement states, in order to generate different connection states with the eight electrode terminals of the said battery interface, so that the different connection states are formed among the first, second, third and fourth strings to output different voltages.

12. The battery pack according to claim 11, wherein said first connection is said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells connected in parallel.

13. The battery pack according to claim 11, wherein said second connection is said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells connected in series parallel.

14. The battery pack according to claim 11, wherein said third connection is said first string of battery cells, said second string of battery cells, said third string of battery cells, and said fourth string of battery cells connected in series.

15. The battery pack according to claim 11, wherein the rated voltage of each string of battery cells is approximately n V, the first rated voltage is n V, the second rated voltage is $2n$ V, the third rated

voltage is 4n V.

16. A battery pack comprising: a plurality of strings of battery cells; a battery interface electrically connected to said plurality of strings of battery cells; said battery interface is configured to output a first rated voltage, a second rated voltage, or a third rated voltage when a respective low voltage plug, a medium voltage plug or a high voltage plug is inserted in said battery interface; when the low voltage plug of a first power tool is inserted in said battery interface, said battery interface is electrically connected to the first power tool, said plurality of strings of battery cells are connected in a first connection; when the medium voltage plug of a second power tool is inserted in said battery interface, said battery interface is electrically connected to the second power tool, said plurality of strings of battery cells are connected in a second connection; when the high voltage plug of a third power tool is inserted in said battery interface, said battery interface is electrically connected to the third power tool, said plurality of strings of battery cells are connected in a third connection; said first connection being different from said second connection and said third connection being different from said first connection and said second connection; each string of the first, second, third and fourth strings of battery cells is provided with a positive electrode and a negative electrode; said battery interface comprises eight electrode terminals each of which is electrically connected to the positive or the negative electrodes of each string, different electric tools are provided with different plugs, and the different plugs is provided with male blades in different arrangement states, in order to generate different connection states with the eight electrode terminals of the said battery interface, so that the different connection states are formed among the first, second, third and fourth strings to output different voltages.
